

Automatic Training Set Expansion for Entity Matching Using LLM-Based Labeling and Web Retrieval

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1 Introduction

Entity Matching, also referred to as entity resolution or record linkage, is the task of identifying records that refer to the same real-world entity across different data sources. It plays a central role in data integration systems, knowledge graph construction, and product aggregation in e-commerce. Recent advances show that transformer-based language models can significantly improve entity matching performance. However, these models require labeled training data consisting of pairs of records annotated as matches or non-matches. Creating such datasets is costly because labels typically require manual annotation. Consequently, many benchmark datasets provide only small training sets to simulate realistic data integration scenarios with limited labeled data.

The goal of this thesis is to investigate methods for automatically extending a small seed training dataset with additional labeled training pairs. Several augmentation strategies will be explored, including string-based augmentation, LLM-assisted labeling of candidate pairs, and web-based retrieval of additional entity descriptions. The usefulness of the generated training data will be evaluated by fine-tuning transformer-based entity matching models and measuring their performance on benchmark datasets.

2 Background

Entity Matching has been extensively studied in the data integration literature. Traditional approaches rely on similarity metrics and rule-based methods to compare attributes such as titles, descriptions, and identifiers. To scale these comparisons to large datasets, blocking techniques are commonly used to restrict comparisons to promising candidate pairs.

Recent work has demonstrated that deep learning methods can significantly improve entity matching performance. In particular, Ditto treats entity matching as a sequence-pair classification problem and fine-tunes pre-trained transformer models such as RoBERTa to determine whether two entity descriptions refer to the same real-world entity. Experimental results show that transformer-based models outperform previous approaches based on manually engineered features and similarity metrics [2].

Although these models achieve strong results, their performance depends heavily on the availability of labeled training data. Creating such datasets is costly because each candidate pair must be manually inspected. Benchmarks such as WDC Products therefore provide evaluation settings with limited training data in order to simulate realistic scenarios in which labeled data is scarce [3].

To address this challenge, several approaches have explored data augmentation techniques for entity matching. For example, Ditto introduces simple text transformations such as token deletion or attribute reordering to generate additional training examples and improve model robustness [1]. More recently, large language models (LLMs) have been investigated for entity matching tasks because they can classify entity pairs and potentially reduce the need for manually labeled training data [3].

Despite these developments, it remains unclear which augmentation strategies produce the most useful additional training data and how automatically generated training pairs influence the performance of transformer-based entity matching models. This thesis aims to investigate this question through a systematic experimental evaluation.

3 Goals and Work Plan

The main goal of this thesis is to investigate methods for automatically extending a small seed training dataset for entity matching and to evaluate the usefulness of the generated training pairs for training transformer-based matching models.

The thesis will address the following research question:

Which automatic data augmentation techniques generate the most useful additional training pairs for entity matching?

To answer this question, several augmentation approaches will be implemented and compared experimentally.

In addition to evaluating model performance, the thesis will analyze the characteristics of the training pairs generated by the different augmentation strategies. Statistics such as overlap between pairs generated by different methods, similarity distributions of entity descriptions, and the proportion of challenging or corner-case pairs will be reported to better understand the composition of the augmented training datasets.

3.1 Task 1 – Literature Review and Familiarization

The first step is to obtain a solid understanding of the relevant literature and tools. This includes studying existing work on entity matching, transformer-based matching models, and data augmentation techniques. Particular attention will be given to the Ditto framework and existing datasets used in entity matching research. The relevant software tools and repositories will also be explored and tested.

Estimated time: 2–3 weeks

3.2 Task 2 – Dataset Preparation and Baseline Model

The thesis will use benchmark datasets commonly used in entity matching research. Possible datasets include WDC Products, Abt–Buy, Walmart–Amazon, and DBLP–Scholar. At least two datasets will be selected to ensure that results generalize across different domains.

A baseline entity matching model will be implemented using existing frameworks such as Ditto or a RoBERTa-based classifier. The model will first be trained using only the original seed training dataset in order to establish a baseline performance.

Estimated time: 3 weeks

3.3 Task 3 – String-Based Data Augmentation

The first augmentation strategy will follow the approach used in the Ditto framework. It applies simple transformations to entity descriptions, such as removing tokens or reordering attributes, to create additional training examples while preserving semantic meaning.

The generated augmented pairs will be added to the training dataset and used to train the matching model again. The resulting model performance will be compared to the baseline.

Estimated time: 3 weeks

3.4 Task 4 – LLM-Based Pair Generation and Labeling

The second approach investigates whether large language models can be used to label additional candidate pairs. Candidate pairs will first be generated using blocking techniques that combine entity descriptions from the seed dataset. An LLM will then classify these pairs as matches or non-matches. The approach may optionally incorporate active learning techniques to prioritize informative candidate pairs.

The automatically labeled pairs will be added to the training dataset, and the matching model will be retrained to evaluate whether the additional data improves performance.

Estimated time: 4 weeks

3.5 Task 5 – Web Retrieval Based Data Augmentation

The third approach explores whether external data from the Web can be used to create additional training examples. For each entity description in the training dataset, web search queries will be generated to retrieve additional offers or entity descriptions. These retrieved descriptions will be paired with the seed entities, and an LLM will be used to label the resulting pairs.

This approach introduces additional real-world descriptions that may improve the robustness of the matching model.

Estimated time: 3 weeks

3.6 Task 6 – Experimental Evaluation and Analysis

The usefulness of the augmented training datasets will be evaluated by fine-tuning entity matching models and measuring their performance on held-out test sets. Standard evaluation metrics such as precision, recall, and F1 score will be used.

The experiments will compare:

- baseline models trained on the original seed dataset
- models trained with string-based augmentation
- models trained with LLM-labeled pairs
- models trained with web-retrieved pairs
- combinations of multiple augmentation strategies

In addition to reporting standard evaluation metrics, the thesis will conduct a qualitative error analysis of the model predictions. Misclassified pairs will be manually inspected and categorized into different error classes (e.g., highly similar non-matching entities, incomplete or noisy entity descriptions, or ambiguous product variants). For each error class, the number of occurrences will be reported in order to better understand the limitations of the models and the influence of the different augmentation strategies.

Estimated time: 3–4 weeks

3.7 Final Phase – Thesis Writing

The final stage will focus on documenting the methodology, experiments, and results in the written thesis.

Estimated time: 3–4 weeks

Table 1: AI tool usage

Task	Tool	Where?
Rephrasing	ChatGPT	Throughout
Sentence Structure	ChatGPT	Throughout

References

- [1] megagonlabs/ditto, February 2026. original-date: 2020-07-13T22:08:15Z.
- [2] Yuliang Li, Jinfeng Li, Yoshihiko Suhara, AnHai Doan, and Wang-Chiew Tan. Deep Entity Matching with Pre-Trained Language Models. *Proceedings of the VLDB Endowment*, 14(1):50–60, September 2020. arXiv:2004.00584 [cs].
- [3] Ralph Peeters, Reng Chiz Der, and Christian Bizer. WDC Products: A Multi-Dimensional Entity Matching Benchmark, June 2023. arXiv:2301.09521 [cs].